Exercițiul 1. (3 puncte)

Pentru $z \in \mathbb{C}$, notăm $z = x + iy$, $x, y \in \mathbb{R}$ şi avem identificarea $z \simeq (x, y)$. Considerăm funcţia $u : \{\mathrm{Re}(z) > 0\} \to \mathbb{R}$, $u(x, y) = x^2 - y^2 + \ln(x^2 + y^2)$. Găsiţi o funcţie olomorfă $f : \{\mathrm{Re}(z) > 0\} \to \mathbb{C}$ pentru care $\mathrm{Re}(f) = u$.

Exercițiul 2. (3 puncte)

Folosind teorema reziduurilor, calculaţi următoarea integrală:

$$\int_{-\infty}^{\infty} \frac{x \cos x}{x^4 + 16} dx.$$

Exercițiul 3. (3 puncte)

Considerăm semi-banda orizontală $L = \{z : z = x + iy, x > 0, y \in (-\pi, 0)\}$. Găsiţi un şir de aplicaţii conforme bijective care transformă L în H^+.

Exercițiul 4. (3 puncte)

Fie $\Omega \subset \mathbb{C}$ un deschis şi $f : \Omega \to \mathbb{C} \setminus \{0\}$ olomorfă. Demonstraţi că $\ln|f|$ este funcţie armonică.

Indicaţie: $\Delta = 4\frac{\partial}{\partial z}\frac{\partial}{\partial \bar{z}}$ şi $\frac{\partial f}{\partial \bar{z}} = \frac{\partial \bar{f}}{\partial z} = 0$ pentru f olomorfă.

1. Mai întâi verificăm dacă u este armonică.

$$\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right)$$

$$\Delta u = \frac{\partial}{\partial x}\left(2x + \frac{2x}{x^2+y^2}\right) + \frac{\partial}{\partial y}\left(-2y + \frac{2y}{x^2+y^2}\right)$$

$\cancel{\Delta u =}$

$$\Delta u = \cancel{2} + \frac{2(x^2+y^2) - 2x\cdot 2x}{(x^2+y^2)^2} + \cancel{(-2)} + \frac{2(x^2+y^2) - 2y\cdot 2y}{(x^2+y^2)^2}$$

$$\Delta u = \frac{2(y^2-x^2)}{(x^2+y^2)^2} + \frac{2(x^2-y^2)}{(x^2+y^2)^2}$$

$\Delta u = 0 \Rightarrow u$ este funcție armonică $\Rightarrow \exists f : \{Re(z)>0\}$ sa olomorfă cu $Re(f) = u$ ($Im(f) = V(x,y)$, $V : \{Re(z)>0\} \to \mathbb{R}$)

$$\frac{\partial f}{\partial z} = \frac{\partial f}{\partial x} = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = \frac{\partial u}{\partial x} - i\frac{\partial u}{\partial y}$$ (conform Relațiilor Cauchy-Riemann)

$$\frac{\partial f}{\partial z} = 2x + \frac{2x}{x^2+y^2} + 2y - \frac{2y}{x^2+y^2}$$

$$\frac{\partial f}{\partial z}(x,0) = 2x + \frac{2}{x} \quad \forall x \in \mathbb{R}, x>0$$

Fie funcția $g : (0,+\infty) \to \mathbb{R}$, $g(x) = x^2 + 2\ln(x)$

$(f-g)'(x) = 0 \quad \forall x \in \mathbb{R}, x>0$; $(0,+\infty)$ are puncte de acumulare în $\mathbb{C}$ $\Rightarrow$ (conform teoremei de identitate a funcțiilor olomorfe) $f \equiv g \Rightarrow f(z) = z^2 + 2\log z \quad \forall z \in \{Re(z)>0\}$

2. $J \overset{not.}{=} \int_{-\infty}^{\infty} \frac{x\cos x}{x^4+16}\,dx$

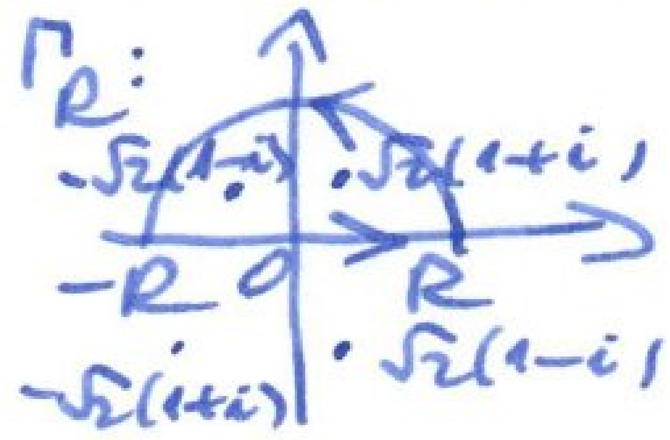
$I \overset{not.}{=} \int_{-\infty}^{\infty} \frac{x\sin x}{x^4+16}\,dx$

$$J+iI=\int_{-\infty}^{\infty}\frac{x(\cos x+i\sin x)}{x^4+16}\,dx=\int_{-\infty}^{\infty}\frac{xe^{ix}}{x^4+16}\,dx$$

Considerăm funcția $f:\Delta\to\mathbb{C}$, $f(z)=\frac{ze^{iz}}{z^4+16}$, unde $\Delta=\mathbb{C}\setminus\{z^4+16=0\}=\mathbb{C}\setminus\{\sqrt2(1+i),-\sqrt2(1+i),\sqrt2(1-i),-\sqrt2(1-i)\}$

$\Rightarrow f$ are 4 poli de ordin 1 ($\sqrt2(1+i),-\sqrt2(1+i),\sqrt2(1-i),-\sqrt2(1-i)$)

Considerăm conturul următor:



Pentru R suficient de mare, doar $\sqrt2(1+i)$ și $-\sqrt2(1-i)$ se află în interiorul conturului Γ_R.

$$Res(f,\sqrt2(1+i))=\lim_{z\to\sqrt2(1+i)}\frac{ze^{iz}}{(z+\sqrt2(1+i))(z-\sqrt2(1-i))(z+\sqrt2(1-i))}$$

$$=\frac{\cancel{\sqrt2(1+i)}}{2\cancel{\sqrt2(1+i)}\cdot2\sqrt2 i\cdot2\sqrt2}\cdot e^{-\sqrt2}(\cos\sqrt2+i\sin\sqrt2)$$

$$=\frac{e^{-\sqrt2}}{8}(-\sin\sqrt2+i\cos\sqrt2)$$

$$Res(f,-\sqrt2(1-i))=\lim_{z\to-\sqrt2(1-i)}\frac{ze^{iz}}{(z-\sqrt2(1+i))(z+\sqrt2(1+i))(z-\sqrt2(1-i))}$$

$$=\frac{\cancel{-\sqrt2(1-i)}}{\cancel{-2\sqrt2(1-i)}\cdot(-2\sqrt2)\cdot2\sqrt2 i}\cdot e^{-\sqrt2}(\cos(-\sqrt2)+i\sin(-\sqrt2))$$

$$=\frac{e^{-\sqrt2}}{8}(\sin\sqrt2+i\cos\sqrt2)$$

Deci, $\int_{\Gamma_R} f(z)\,dz=\frac{e^{-\sqrt2}}{4}\cdot i\cos\sqrt2$

Acum, scriem integrala și în alt mod, parametrizând curba Γ_R pe cele 2 porțiuni netede, după care facem $R \to +\infty$.

$$\int_{\Gamma_R} f(z)\,dz = \int_{-R}^{R} \frac{x e^{ix}}{x^4+16}\,dx + \int_0^{\pi} \frac{Re^{it} \cdot e^{iRe^{it}}}{(Re^{it})^4+16}\,dt$$

$$\left| \int_0^{\pi} \frac{Re^{it} \cdot e^{iRe^{it}}}{(Re^{it})^4+16}\,dt \right| \le \int_0^{\pi} \left| \frac{Re^{it} \cdot e^{iRe^{it}}}{(Re^{it})^4+16} \right| dt \le$$

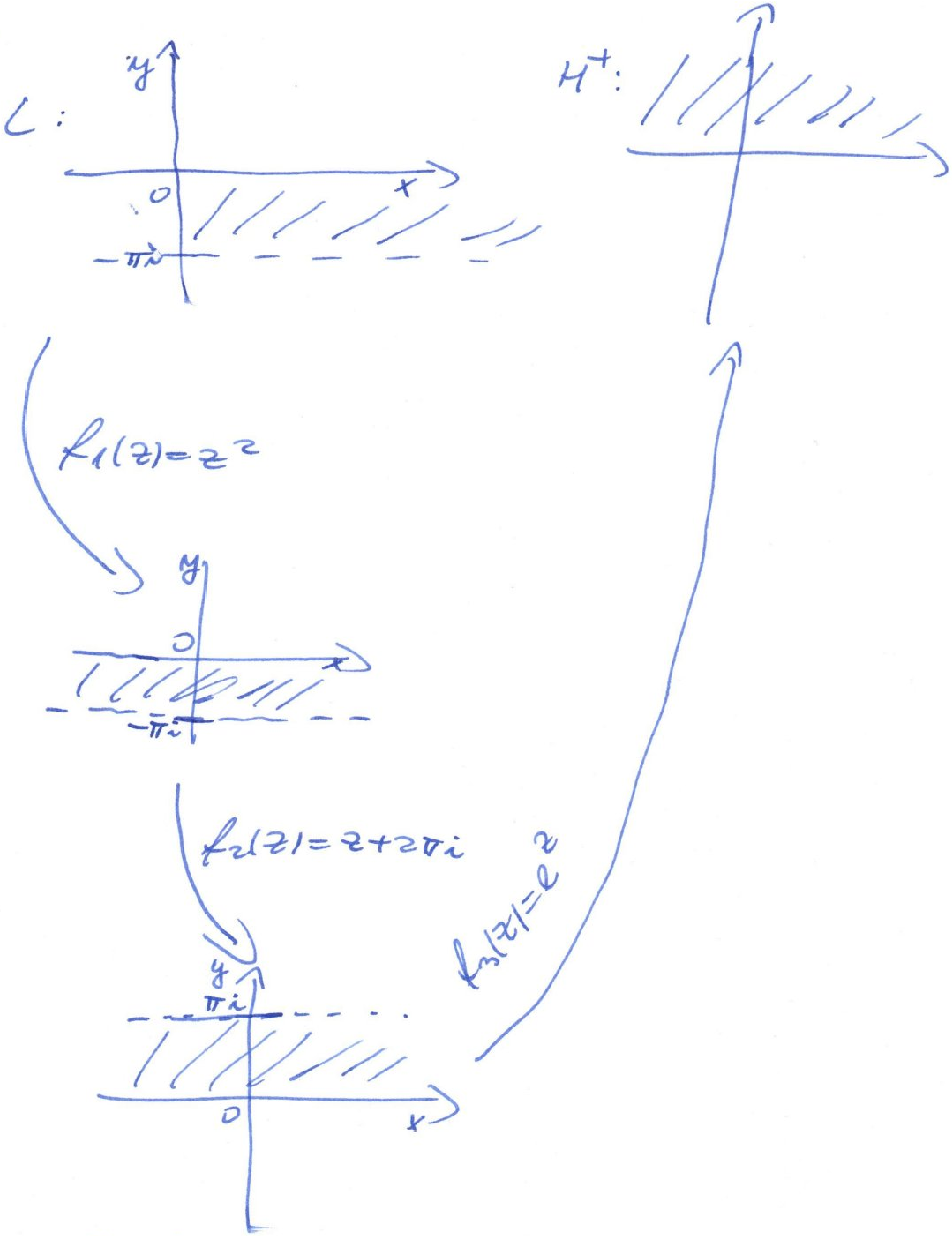
$$\le \int_0^{\pi} \frac{R}{R^4-16}\,dt = \frac{R\pi}{R^4-16} \xrightarrow{R \to +\infty} 0$$

$$\int_{-R}^{R} \frac{x e^{ix}}{x^4+16}\,dx \xrightarrow{R \to +\infty} \int_{-\infty}^{\infty} \frac{x e^{ix}}{x^4+16}\,dx = J + iI$$

În concluzie, $J + iI = \int_{\Gamma_R} f(z)\,dz = \frac{e^{-\sqrt{2}}}{4} \cdot i \cos\sqrt{2}$

$\Rightarrow J = 0.$

3.



$f=f_3\circ f_2\circ f_1$

$f=f_3(f_2(f_1(z)))$

$f=f_3(f_1(z)+2\pi i)$

$f=e^{f_1(z)}$

$f=e^{z^2}$

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4. $f = u(x,y) + iv(x,y)$, unde $u, v : \Omega \to \mathbb{R}$, u, v armonice și satisfac Relațiile Cauchy – Riemann

$$\ln|f| = \ln\left(\sqrt{u^2(x,y)+v^2(x,y)}\right)$$

$$\cdot \Delta \ln|f| = \frac{\partial}{\partial x}\left(\frac{\partial \ln|f|}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{\partial \ln|f|}{\partial y}\right)$$

$$\frac{\partial \ln|f|}{\partial x} = \frac{1}{\sqrt{u^2(x,y)+v^2(x,y)}} \cdot \frac{\cancel{2}u(x,y)\cdot\frac{\partial u}{\partial x}(x,y) + \cancel{2}v(x,y)\cdot\frac{\partial v}{\partial x}(x,y)}{\cancel{2}\sqrt{u^2(x,y)+v^2(x,y)}}$$

$$= \frac{u(x,y)\cdot\frac{\partial u}{\partial x}(x,y) + v(x,y)\frac{\partial v}{\partial x}(x,y)}{u^2(x,y)+v^2(x,y)}$$

$$\frac{\partial}{\partial x}\left(\frac{\partial \ln|f|}{\partial x}\right) = \Big[\left(u^2(x,y)+v^2(x,y)\right)\Big(u(x,y)\cdot\frac{\partial^2 u}{\partial x^2}(x,y) + \left(\frac{\partial u}{\partial x}(x,y)\right)^2 +$$

$$+ v(x,y)\cdot\frac{\partial^2 v}{\partial x^2}(x,y) + \left(\frac{\partial v}{\partial x}(x,y)\right)^2\Big) - \left(u(x,y)\cdot\frac{\partial u}{\partial x}(x,y) + v(x,y)\cdot\frac{\partial v}{\partial x}(x,y)\right)$$

$$\cdot 2\left(u(x,y)\cdot\frac{\partial u}{\partial x}(x,y) + v(x,y)\cdot\frac{\partial v}{\partial x}(x,y)\right)\Big] : \left[u^2(x,y)+v^2(x,y)\right]^2$$

$$= \Big[u^3(x,y)\cdot\frac{\partial^2 u}{\partial x^2}(x,y) + v^2(x,y)\,u(x,y)\frac{\partial^2 u}{\partial x^2}(x,y)$$

$$+ u^2(x,y)v(x,y)\frac{\partial^2 v}{\partial x^2}(x,y) + \frac{\partial^2 v}{\partial x^2}(x,y)\cdot v^3(x,y) + u^2(x,y)\left(\frac{\partial u}{\partial x}(x,y)\right)^2$$

$$+ v^2(x,y)\cdot\left(\frac{\partial u}{\partial x}(x,y)\right)^2 + u^2(x,y)\left(\frac{\partial v}{\partial x}(x,y)\right)^2 + v^2(x,y)\left(\frac{\partial v}{\partial x}(x,y)\right)^2$$

$$- 2u^2(x,y)\cdot\left(\frac{\partial u}{\partial x}(x,y)\right)^2 - 2v^2(x,y)\cdot\left(\frac{\partial v}{\partial x}(x,y)\right)^2 - 4u(x,y)v(x,y)\frac{\partial u}{\partial x}(x,y)$$

$$\cdot\frac{\partial v}{\partial x}(x,y)\Big] : \left[u^2(x,y)+v^2(x,y)\right]^2$$

$$= \Big[u^3(x,y)\cdot\frac{\partial^2 u}{\partial x^2}(x,y) + v^2(x,y)u(x,y)\frac{\partial^2 u}{\partial x^2}(x,y) + v^3(x,y)\cdot\frac{\partial^2 v}{\partial x^2}(x,y)$$

$$+ v(x,y)u^2(x,y)\frac{\partial^2 v}{\partial x^2}(x,y) - u^2(x,y)\cdot\left(\frac{\partial u}{\partial x}(x,y)\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2\left(u^2(x,y)\right) +$$

$$+ 4u(x,y)v(x,y)\frac{\partial u}{\partial x}(x,y)\frac{\partial u}{\partial y}(x,y)\Big] : \left[u^2(x,y)+v^2(x,y)\right]^2$$

Analog, $\frac{\partial^2 \ln|f|}{\partial y^2} = \Big[u^3(x,y)\cdot\frac{\partial^2 u}{\partial y^2}(x,y) + v^2(x,y)u(x,y)\frac{\partial^2 u}{\partial y^2}(x,y) + v^3(x,y)$

$$\frac{\partial^2 v}{\partial y^2}(x,y) + v(x,y)u^2(x,y)\cdot\frac{\partial^2 v}{\partial y^2}(x,y) + u^2(x,y)\cdot\left(\frac{\partial u}{\partial x}(x,y)\right)^2 - \left(\frac{\partial u}{\partial y}(x,y)\right)^2\cdot u^2(x,y)$$

$$-4u(x,y)v(x,y)\frac{\partial u}{\partial x}(x,y)\frac{\partial u}{\partial y}(x,y)\Big] : \left[u^2(x,y)+v^2(x,y)\right]^2$$

În concluzie, $\Delta \ln|f| = 0 \Rightarrow \ln|f|$ este funcție armonică